

Study Questions: Systems — Reading the Pattern as a Whole

1. Pick a crochet pattern you have used recently. What are its **prior beliefs** — the assumptions it makes about yarn, hook size, gauge, and your skill level?
2. How does a gauge swatch function as **model validation**? What happens to the system's predictions if you skip the gauge swatch?
3. Explain why the stitch count at the end of a row functions as a **checksum**. What information does it give you about the state of the system?
4. In what sense is the crochet pattern a **Markov blanket** — a boundary between the designer's mind and your hands?
5. Compare a one-page free pattern with no photos to a 20-page tech-edited pattern with schematics. How do their system boundaries differ in terms of information richness?
6. Why can two crocheters follow the exact same pattern and produce noticeably different finished objects? Frame your answer in terms of prediction error.
7. A pattern says "worsted weight yarn, any brand." How does this vague prior specification affect the system's ability to predict the finished object?
8. When a pattern lists "special stitches" with definitions at the top, what role does this section play in the generative model?
9. How would you describe the difference between a written pattern and a video tutorial in terms of the system boundary? Which provides a richer sensory interface?
10. A pattern calls for a 5.0mm hook but your gauge swatch is too loose. You switch to a 4.5mm hook. In systems terms, what have you just done?
11. Why do experienced crocheters sometimes say they can "tell" a pattern will not work just by reading it? What system-level information are they detecting?

12. A pattern includes a schematic diagram showing finished measurements. How does this schematic function differently from the row-by-row instructions as a component of the generative model?
13. Consider a modular blanket pattern where each square is made separately and then joined. How does the system structure differ from a pattern worked in one piece?
14. What does it mean for a pattern to be "well-tested"? Translate this into Active Inference language about model accuracy and prediction error.
15. A pattern says "repeat rows 3-6 until piece measures 12 inches." Why is this instruction less precise as a system specification than "repeat rows 3-6 a total of 15 times"?
16. How does the materials list in a pattern function differently from the stitch instructions? Map each to its role in the generative model.
17. When you substitute a different yarn than what the pattern calls for, you are changing one of the system's priors. What downstream effects might this have on the posterior (finished object)?
18. Explain how a crochet pattern for a garment with size options (S, M, L, XL) is actually encoding multiple generative models within a single document.
19. A crocheter posts on Ravelry: "I followed the pattern exactly but my hat was too small." Analyze this situation using the concepts of prior beliefs, prediction error, and model validation.
20. If you were to write a crochet pattern from scratch, what would you include to minimize prediction error for the person reading it? Design the thickest possible Markov blanket.